Maths Exploring the Terminal Assessment-09

QCM A · scalar product and orthogonality

Orthonormal coordinate system. After S51; QCM B after S56.

For each question, only one proposal is accurate.
consult the correction.

1. $(1;2;-1)\cdot(2;0;3)$ is valid for:

A. 5
B. -1
C. 1
D. -3

2. The norm of $(2;-1;2)$ is:

A. 9
B. 5
C. $\sqrt{5}$
D. 3

3. To prove that a vector is normal to a plane, simply test its orthogonality at:

A. Two non-collinear directions of the plan
B. a direction of the plan
C. Two colinear directions of the plan
D. a normal plan

4. Two non-zero vectors of negative scalar product form an angle:

A. Acute
B. Law
C. obtuse
D. None

5. Two orthogonal lines of space:

A. are still stinging
B. may be non-coplanar
C. are parallel
D. are confused

MCQ B · projection, minimum and volume

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consult the correction.

6. The project $M(2;3;4)$ on the Plan $z=0$ is:

A. $(0;0;4)$
B. $(2;3;4)$
C. $(0;0;0)$
D. $(2;3;0)$

7. H is the project of M on a plan. P of the plan:

A. $MP^2=MH^2+HP^2$
B. $MP=MH+HP$
C. $MH^2=MP^2+HP^2$
D. $MP=HP$

8. A base tetrahedron of floor 6 and height 4 has for volume:

A. 24
B. 12
C. 8
D. 6

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9. For vectors u, v units such as $u \cdot v = -\frac{1}{2}$, the angle is:

A. $\pi/3$ B. $2\pi/3$

C. $\pi/2$ D. π

10. $A(1;0;0)$, $M(0;2;3)$, and d of Director $(1;1;0)$ passing through A. The planned M on d is:

A. $(1;1;0)$ B. $(0;2;0)$

C. $(1;0;0)$ D. $(\frac{3}{2};\frac{1}{2};0)$

True or false · expected justification. For each statement, indicate whether it is true or false, and then write evidence or a counter-example.

1. Any point in a plan is closer to M that his project.

2. Two vectors of zero scalar product are colinear.

3. Calculate a volume from two different sides shall give the same result.